Products Carbon Footprint Report 2024 Relats

1. Introduction

This report presents the results of a Product Carbon Footprint (PCF) assessment conducted by Footprint Mappa for Relats for the year 2024. The primary objective of this assessment is to quantify the greenhouse gas (GHG) emissions associated with the cradle-to-gate life cycle of the selected products, encompassing three distinct stages: materials acquisition, manufacturing, and transport. By establishing a comprehensive emissions inventory across these life cycle stages, this assessment provides a foundation for understanding the climate impact of Relats' product portfolio.

The purpose of this document is to provide Relats with a clear and transparent overview of each product's climate impact, enabling the identification of the most emission-intensive life cycle stages and supporting informed decision-making regarding material optimisation, process improvements, and sustainable product design initiatives. By quantifying emissions across the value chain, this assessment equips the organisation with the insights necessary to develop targeted reduction strategies and enhance environmental performance.

Relats is an organisation whose operations and product manufacturing processes are covered within the assessment boundary of this study. The company's production activities, supply chain management, and manufacturing facilities form the basis of the data collection and analysis presented herein.

This analysis covers all products manufactured by Relats during the 2024 calendar year. As this represents the baseline PCF assessment for the organisation, results have not been benchmarked against prior years; however, this assessment establishes a reference point for tracking future emissions reductions and performance improvements. The assessment was performed in accordance with ISO 14067:2018, the international standard for quantifying and reporting the carbon footprint of products, thereby ensuring methodological consistency and credibility across all assessed products.

2. Methodological Approach

The 2024 PCF assessment for Relats has been carried out in accordance with internationally recognised standards: ISO 14067:2018 – Greenhouse gases – Carbon footprint of products; GHG Protocol – Product Life Cycle Accounting and Reporting Standard; and life-cycle inventory (LCI) data from reputable databases such as EXIOBASE, DEFRA, IEA, and OCCC. These methodological foundations ensure robustness, comparability, and auditability of results across all assessed products.

To ensure robustness and transparency, the calculation is based on a combination of primary and secondary data. Primary data were collected directly from the company's internal sources, including material compositions, energy consumption per process, production yields, packaging specifications and transport distances. Secondary data were drawn from internationally recognised emission factor databases, such as DEFRA, IEA, OCCC, and EXIOBASE, depending on the category and geographical location.

3. Scope and Boundaries

This assessment encompasses a comprehensive Product Carbon Footprint evaluation of six products conducted under a unified operational boundary. The system boundary adopted follows a cradle-to-gate approach in accordance with ISO 14067, encompassing the life cycle stages from materials acquisition through manufacturing and transportation to the factory gate. This delineation captures the environmental impacts associated with raw material extraction, production processes, and logistics up to the point of product delivery, while excluding subsequent distribution, use phase, and end-of-life stages. The assessment applies consistent methodological parameters across all six products to enable meaningful comparative analysis within the defined scope, notwithstanding the absence of a geographically specified manufacturing location.

Product
Botella PET 500ml
Caja carton corrugado
Camiseta algodon
Panel solar 400W
Silla de oficina
Detergente liquido 1L

The boundaries defined for this PCF study include the following life-cycle stages:

- **Raw material and packaging acquisition:** polymers, fibres, resins and all material components.
- **Upstream transportation:** delivery of materials, packagings and consumables to production sites.
- **Energy and consumables used for manufacturing processes:** including thermal, mechanical and chemical operations.
- **Waste generated during manufacturing:** treatment and final disposal of production scraps.
- Downstream transportation of waste to adequate manager.

Material Acquisition	1.1 Raw materials 1.2 Inbound packaging 1.3 Outbound packaging
Manufacturing	2.1 Electricity 2.2 Other energy 2.3 Consumables 2.4 Waste generated
Transportation	3.1 Upstream transport of raw materials and packaging 3.2 Upstream transport of consumables and additives

	3.3 Downstream transport of waste to waste manager
	3.4 Internal transport between sites
Distribution	4.1 Product distribution
Use	5.1 Product use
	5.2 Maintenance and servicing
	5.3 Other use-stage emissions
End-of-life	6.1 Collection and transport of end-of-life products
	6.2 End-of-life treatment
	6.3 Final disposal

3.1 Botella PET 500ml

This section presents the production route and carbon footprint results for Botella PET 500ml.

Material Acquisition	Manufacturing	Transportation
Distribution	Use	End-of-life

3.2 Caja carton corrugado

This section presents the production route and carbon footprint results for Caja carton corrugado.

Material Acquisition	Manufacturing	Transportation
Distribution	Use	End-of-life

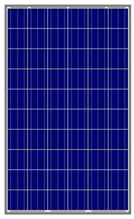
3.3 Camiseta algodon

This section presents the production route and carbon footprint results for Camiseta algodon.

Material Acquisition	Manufacturing	Transportation
Distribution	Use	End-of-life

3.4 Panel solar 400W

This section presents the production route and carbon footprint results for Panel solar 400W.



Material Acquisition	Manufacturing	Transportation
Distribution	Use	End-of-life

3.5 Silla de oficina

This section presents the production route and carbon footprint results for Silla de oficina.

Material Acquisition	Manufacturing	Transportation
Distribution	Use	End-of-life

3.6 Detergente liquido 1L

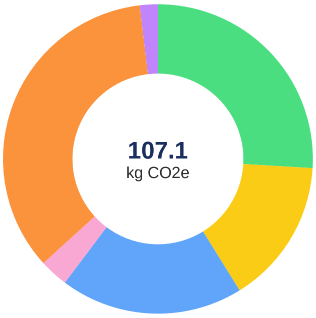
This section presents the production route and carbon footprint results for Detergente liquido 1L.

Material Acquisition	Manufacturing	Transportation
Distribution	Use	End-of-life

4. Results

4.1 Botella PET 500ml

The Botella PET 500ml product presents a total carbon footprint of 107.075 kg CO2e per functional unit.

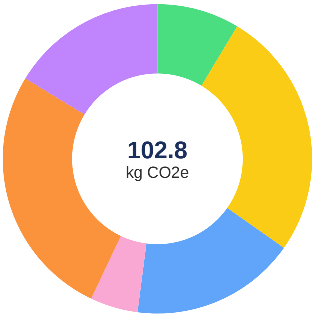


- **Material Acquisition**: 27.781 kg CO2e (25.9%)
- **Manufacturing**: 16.290 kg CO2e (15.2%)
- **Transportation**: 20.450 kg CO2e (19.1%)
- **Distribution**: 3.201 kg CO2e (3.0%)
- **Use**: 37.345 kg CO2e (34.9%)
- **End-of-life**: 2.008 kg CO2e (1.9%)

The Use phase represents the largest contributor to the Botella PET 500ml carbon footprint, accounting for approximately 34.9% of total emissions.

4.2 Caja carton corrugado

The Caja carton corrugado product presents a total carbon footprint of 102.775 kg CO2e per functional unit.

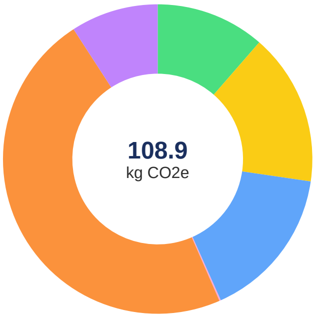


- **Material Acquisition**: 8.865 kg CO2e (8.6%)
- **Manufacturing**: 26.864 kg CO2e (26.1%)
- **Transportation**: 17.770 kg CO2e (17.3%)
- **Distribution**: 5.131 kg CO2e (5.0%)
- **Use**: 27.392 kg CO2e (26.7%)
- **End-of-life**: 16.753 kg CO2e (16.3%)

The Use phase represents the largest contributor to the Caja carton corrugado carbon footprint, accounting for approximately 26.7% of total emissions.

4.3 Camiseta algodon

The Camiseta algodon product presents a total carbon footprint of 108.876 kg CO2e per functional unit.

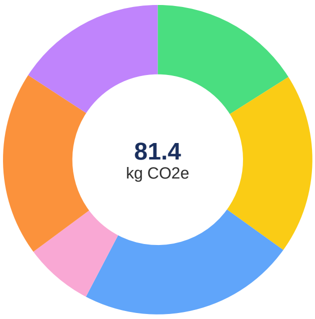


- **Material Acquisition**: 12.416 kg CO2e (11.4%)
- **Manufacturing**: 17.324 kg CO2e (15.9%)
- **Transportation**: 17.414 kg CO2e (16.0%)
- **Distribution**: 0.166 kg CO2e (0.2%)
- **Use**: 51.591 kg CO2e (47.4%)
- **End-of-life**: 9.965 kg CO2e (9.2%)

The Use phase represents the largest contributor to the Camiseta algodon carbon footprint, accounting for approximately 47.4% of total emissions.

4.4 Panel solar 400W

The Panel solar 400W product presents a total carbon footprint of 81.382 kg CO2e per functional unit.

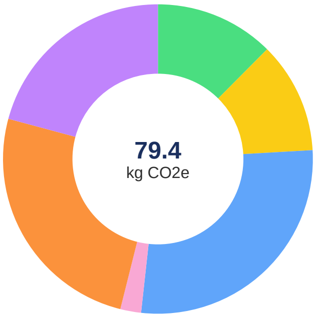


- **Material Acquisition**: 13.043 kg CO2e (16.0%)
- **Manufacturing**: 15.376 kg CO2e (18.9%)
- **Transportation**: 18.521 kg CO2e (22.8%)
- **Distribution**: 5.842 kg CO2e (7.2%)
- **Use**: 15.742 kg CO2e (19.3%)
- **End-of-life**: 12.858 kg CO2e (15.8%)

The Transportation phase represents the largest contributor to the Panel solar 400W carbon footprint, accounting for approximately 22.8% of total emissions.

4.5 Silla de oficina

The Silla de oficina product presents a total carbon footprint of 79.419 kg CO2e per functional unit.

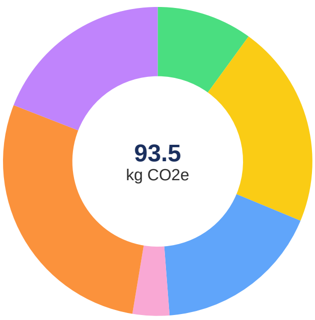


- **Material Acquisition**: 9.886 kg CO2e (12.4%)
- **Manufacturing**: 9.239 kg CO2e (11.6%)
- **Transportation**: 21.965 kg CO2e (27.7%)
- **Distribution**: 1.722 kg CO2e (2.2%)
- **Use**: 20.079 kg CO2e (25.3%)
- **End-of-life**: 16.528 kg CO2e (20.8%)

The Transportation phase represents the largest contributor to the Silla de oficina carbon footprint, accounting for approximately 27.7% of total emissions.

4.6 Detergente liquido 1L

The Detergente liquido 1L product presents a total carbon footprint of 93.518 kg CO2e per functional unit.



- **Material Acquisition**: 9.350 kg CO2e (10.0%)
- **Manufacturing**: 19.894 kg CO2e (21.3%)
- **Transportation**: 16.370 kg CO2e (17.5%)
- **Distribution**: 3.594 kg CO2e (3.8%)
- **Use**: 26.480 kg CO2e (28.3%)
- **End-of-life**: 17.830 kg CO2e (19.1%)

The Use phase represents the largest contributor to the Detergente liquido 1L carbon footprint, accounting for approximately 28.3% of total emissions.

5. Strategic Recommendations

Footprint Mappa has identified the following main lines to decarbonise the products assessed for Relats.

5.1 Prioritize Use-Phase Decarbonization for Consumer Products

The use phase represents the largest emissions contributor for both the Botella PET 500ml (37.345 kg CO2e, 34.9% of total) and Camiseta algodón (51.591 kg CO2e, 47.4% of total). For the PET bottle, this reflects refrigeration and consumer handling emissions, while for the cotton shirt, it represents washing and drying impacts. Relats should implement product design interventions such as optimizing packaging for reduced refrigeration requirements, developing care labels that encourage cold-water washing, and exploring material modifications that reduce water absorption and drying time. These measures can deliver substantial emissions reductions without requiring supply chain restructuring.

5.2 Optimize Manufacturing Processes Across All Product Lines

Manufacturing phase emissions are disproportionately high for the Caja cartón corrugado (26.864 kg CO2e, 26.1% of total), suggesting inefficiencies in production energy intensity. Combined with elevated manufacturing emissions in the Detergente líquido 1L (19.894 kg CO2e, 21.3% of total) and Botella PET 500ml (16.290 kg CO2e, 15.2% of total), this indicates systematic opportunities for operational improvement. Relats should conduct energy audits of manufacturing facilities, transition to renewable electricity sources where feasible, upgrade equipment to higher-efficiency machinery, and implement lean manufacturing protocols to reduce waste and energy consumption per unit produced.

5.3 Reduce Material Acquisition Emissions Through Supplier Engagement

Material acquisition represents the largest single-phase contribution for the Botella PET 500ml (27.781 kg CO2e, 25.9% of total), driven by virgin plastic production and energy-intensive refining processes. Relats should prioritize sourcing recycled PET from suppliers with documented lower-carbon extraction and processing methods, while simultaneously investing in closed-loop recycling partnerships to increase the proportion of recycled content in future bottle productions. Parallel engagement with cotton suppliers for the Camiseta algodón should focus on regenerative agriculture practices and organic certification, which typically reduce material acquisition emissions by 15-30% compared to conventional sourcing.

5.4 Streamline Distribution Networks and Transportation Routes

Transportation and distribution collectively account for significant emissions across multiple products—particularly the Silla de oficina (23.687 kg CO2e combined, 29.8% of total) and Botella PET 500ml (20.450 kg CO2e transportation alone, 19.1% of total). Relats should consolidate shipments, optimize logistics routes using AI-powered planning tools, and transition carrier partnerships toward lower-emission options including rail freight where feasible and electric vehicle fleets for last-mile delivery. For heavier products like office furniture, exploring local or regional manufacturing partnerships can substantially reduce long-distance transportation requirements and associated emissions.